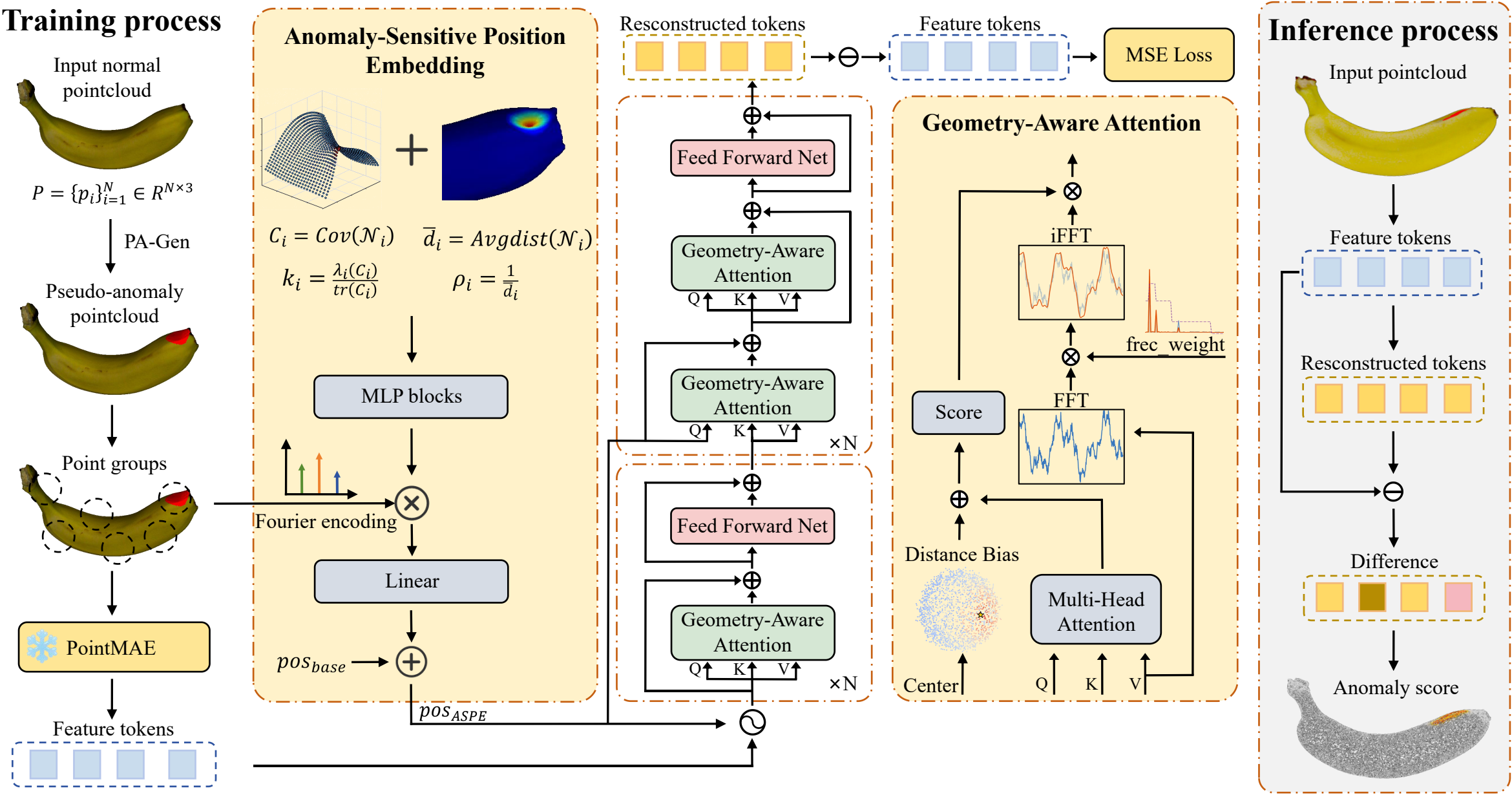

Training process
Input normal pointcloud
$P = \{p_i\}_{i=1}^{N} \in R^{N\times 3}$
PA-Gen
Pseudo-anomaly pointcloud
Point groups
PointMAE
Feature tokens
Anomaly-Sensitive Position Embedding
$C_i = Cov(\mathcal{N}_i)$
$\overline{d}_i = Avgdist(\mathcal{N}_i)$
$k_i = \frac{\lambda_i(C_i)}{tr(C_i)}$
$\rho_i = \frac{1}{\overline{d}_i}$
MLP blocks
Fourier encoding
Linear
pos_{base}
pos_{ASPE}
Resconstructed tokens
Feature tokens
MSE Loss
Feed Forward Net
Geometry-Aware Attention
Q
K
V
×N
Geometry-Aware Attention
iFFT
frec_weight
FFT
Score
Distance Bias
Center
Multi-Head Attention
Inference process
Input pointcloud
Feature tokens
Resconstructed tokens
Difference
Anomaly score